MIE 100 – LEC 0103

Lecture 4

2.4 Rectangular Coordinates (x-y)

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Chapter 2: Kinematics of Particles

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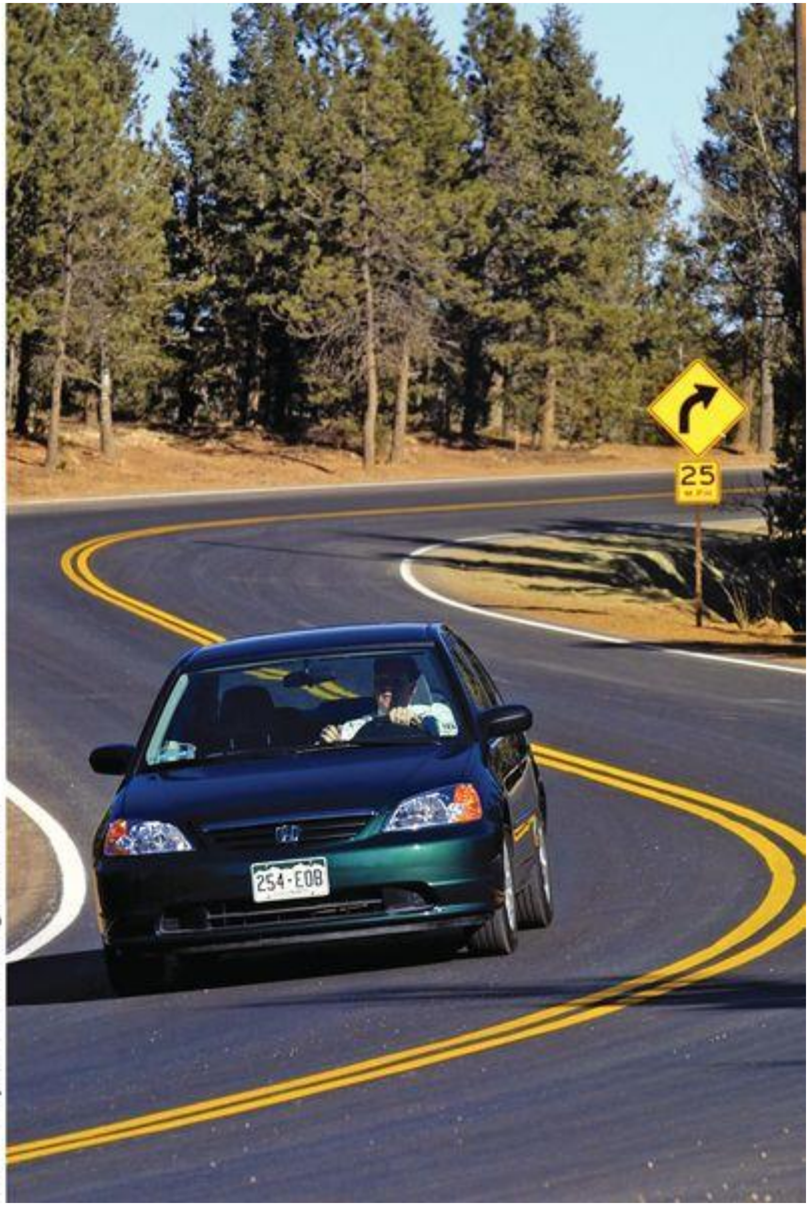


Sean Cayton/The Image Works

- **2.1 Introduction**
- **2.2 Rectilinear Motion**
- **2.3 Plane Curvilinear Motion**
- **2.4 Rectangular Coordinates (x-y)**
- **2.5 Normal and Tangential Coordinates (n-t)**
- **2.6. Polar Coordinates (r-θ)**
- 2.8 Relative Motion (Translating Axes)
- 2.9 Constrained Motion of Connected Particles

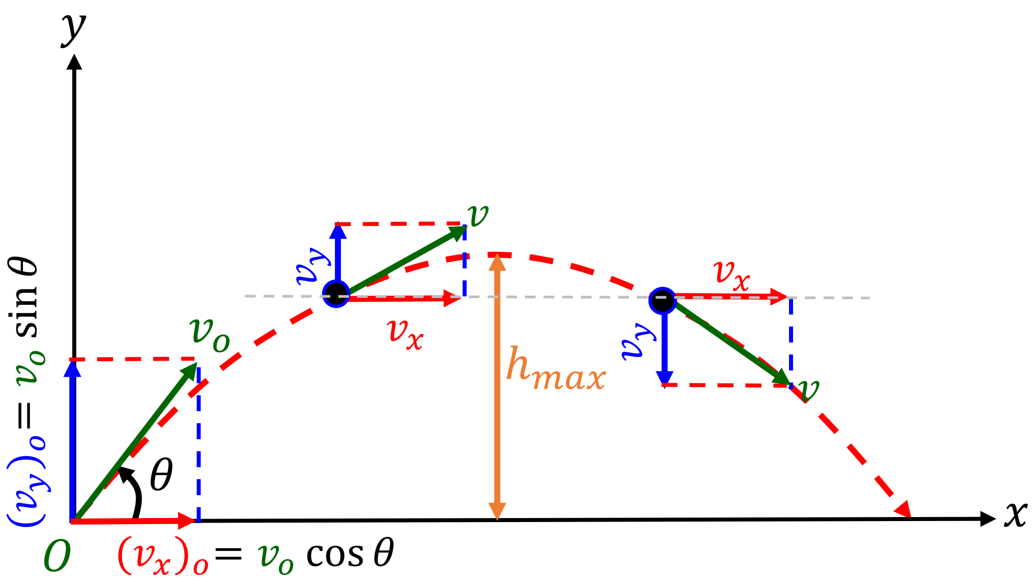
The puzzle

Learning Objective

In this lecture, you will learn to

- Apply the motion equations for analyzing projectile motion
- Analyze the free-flight motion of a projectile particle

Rectangular Coordinates: Projectile Motion



- Projectile motion can be treated as two-dimensional rectilinear motions
- Horizontal direction
  - v_x is constant, therefore acceleration is zero

$$a_x = \frac{dv_x}{dt} = 0$$

- Vertical direction
  - Acceleration is constant (i.e., gravity):

$$a_y = \frac{dv_y}{dt} = a_c = -g$$

  - At maximum height, h_{max},

$$v_y = 0$$

Assumptions
- Neglect aerodynamic drag
- Neglect curvature and rotation of earth
- Gravity is constant

Projectile Motion: Velocity and Position

Horizontal motion, $\vec{a}_x = 0$:

- Velocity

$$v_x = (v_x)_o$$

- Position

$$x = x_o + (v_x)_o t$$

Vertical motion, $\vec{a}_y = -g$:

- Velocity

$$v_y = (v_y)_o - gt$$

- Position

$$y = y_o + (v_y)_o t - 1/2(gt^2)$$

- Velocity-Position

$$v_y^2 = (v_y)_0^2 - 2g(y - y_o)$$

Recall: Constant Acceleration Eqns.

$$a_c = \frac{dV}{dt} = V\frac{dV}{ds}$$

Velocity:

$$V = V_o + a_c t$$

$$V^2 = V_o{}^2 + 2a_c(s - s_o)$$

Position:

$$s = s_o + V_o t + \frac{a_c t^2}{2}$$

where s_o and V_o are the position and velocity at the initial time, t = 0

Example Problem [2.4-1]

Given: v_o and θ , R

Find: The equation that defines y as a function of x

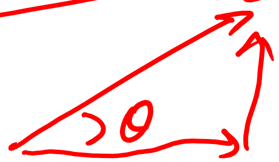
Assumptions:

no wind drag $a_x = 0$

g = const. $a = -g$

At origin O, $x_0 = y_0 = 0$

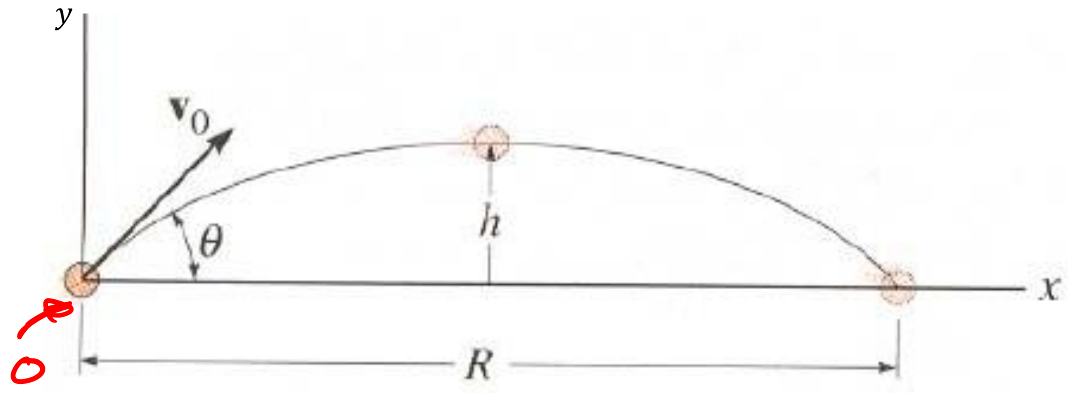
@ $t=0$. $\vec{V}_0$



Equations:

$\cos\theta = \frac{V_{x,0}}{V_0}$, $V_{x,0} = V_0 \cos\theta$

$\sin\theta = \frac{V_{y,0}}{V_0}$, $V_{y,0} = V_0 \sin\theta$



Special cases
$(a = 0)\quad s = s_0 + v_0 t$
$(\text{const a})\quad v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$

Solutions:

x-dir^n: (1) →

$x = \cancel{x_0} + (\underbrace{v_{x,0}}_{})t$

$x = (\underline{V_0 \cos\theta})t$

$t = \dfrac{x}{V_0 \cos\theta}$ — (5)

y-dir^n: (3) → $a = -g$

$y = 0 + V_{y,0}t - \frac{1}{2}gt^2$ — (6)

Sub. 5 → (6)

$y = V_{y,0}\left(\dfrac{x}{V_0\cos\theta}\right) - \frac{1}{2}g\left(\dfrac{x}{V_0\cos\theta}\right)^2$

Trig rules:

$$tan\theta = \frac{sin\theta}{cos\theta}$$

$$\underline{\tan^2\theta + 1} = \sec^2\theta = \underline{1/\cos^2\theta}$$ (3)

Special cases
(1) (a = 0) $s = s_0 + v_0 t$
(2) (const a) $v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2}a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$ (4)

Sub. $V_{y,0}$

$y = (V_0 \sin\theta)\left(\dfrac{x}{V_0\cos\theta}\right) - \frac{1}{2}g\left(\dfrac{x^2}{V_0^2\cos^2\theta}\right)$

$$\boxed{y = x\tan\theta - \frac{1}{2}\frac{gx^2}{V_0^2}(1+\tan^2\theta)}$$

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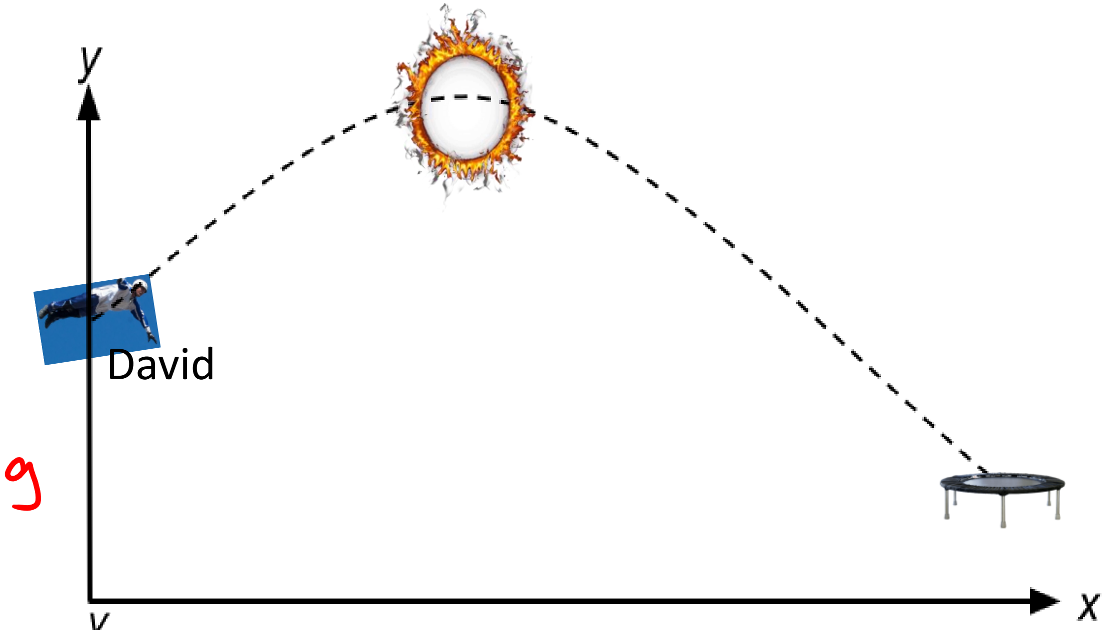
Example Problem [2.4-2]

Given: Initial velocity v_o, launch angle θ

Find: **(i)** The horizontal location for placing the net

(ii) Vertical and horizontal location for ring of fire

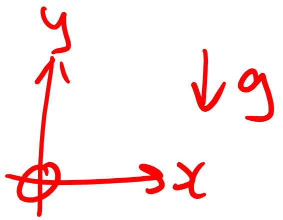


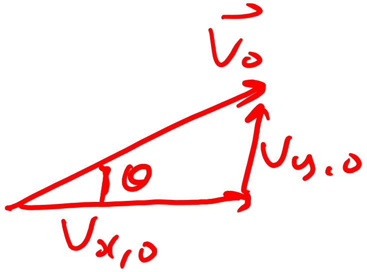
Assumptions & Initial Conditions:

no wind drag $\quad a_x = 0$

$a = -g$

$$y_o = 10m$$
$$y_1 = 5m$$
$$v_o = 10\frac{m}{s}$$
$$\theta = 30^o$$



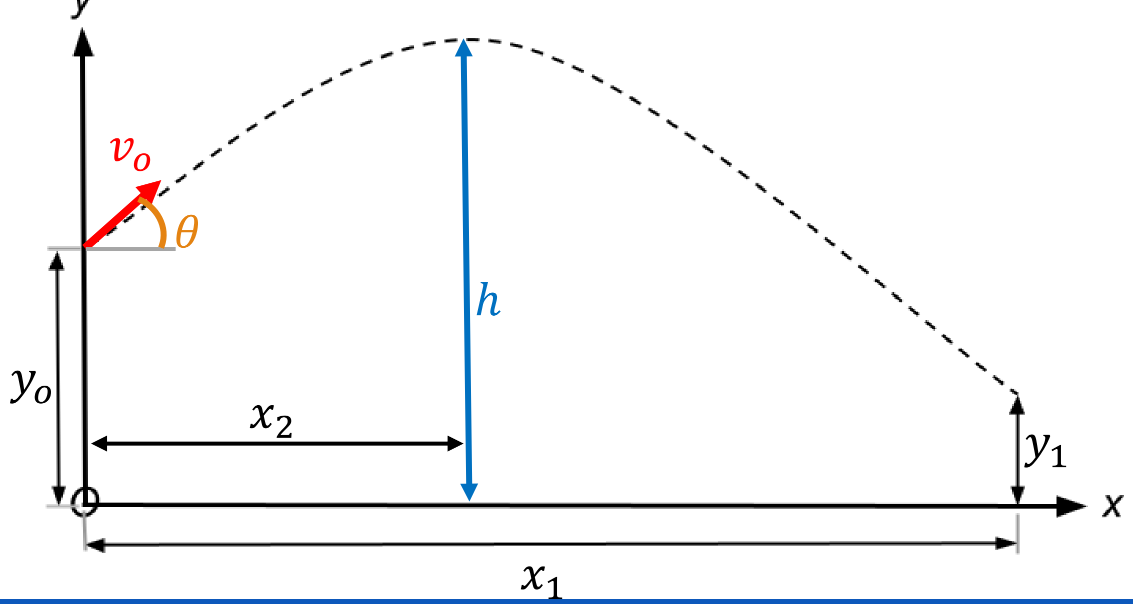


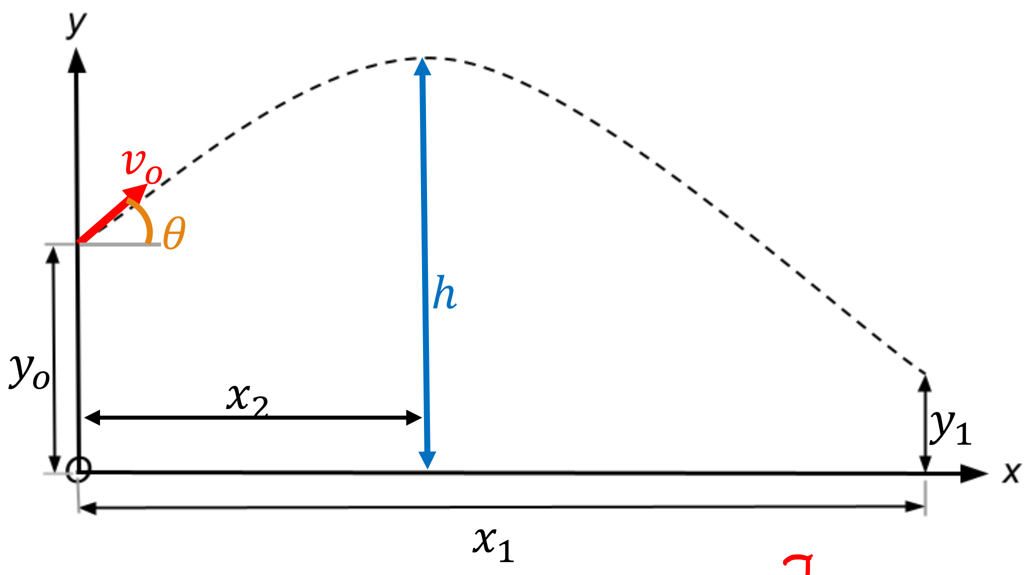
Equations:

x-dirⁿ : $v_{x,0} = \vec{v}_o \cos\theta$

y-dirⁿ : $v_{y,0} = \vec{v}_o \sin\theta$

Special cases
(a = 0) $\quad s = s_0 + v_0 t$
(const a) $\quad v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$





Solutions (i):

x-dirⁿ: (1) → $x_1 = \cancel{x_0}^{\,0} + (v_{x,0})t$

$x_1 = (v_0 \cos\theta)t$ —— (5)

y-dirⁿ: (3) → $y = y_0 + (v_{y,0})t + \frac{1}{2}at^2$

$y_1 = y_0 + (v_0 \sin\theta)t - \frac{g}{2}t^2$ —— (6)

I.C.

$y_o = 10m$
$y_1 = 5m$
$v_o = 10\frac{m}{s}$
$\theta = 30^o$

Sub. I.C. into (6),

$5 = 10 + (10 \sin 30^\circ)t - \frac{9.81}{2}t^2$

$4.905t^2 - 5t - 5 = 0$

$t = 1.64s$ ←

or

$-0.62s$ (rejected)

Sub. t=1.64 into (5)

$x_1 = (10\cos 30^\circ)(1.64)$

$\boxed{= 14.2m}$

Special cases	
(1) —	$(a = 0) \quad s = s_0 + v_0 t$
(2) —	(const a) $v = v_0 + a_0 t$
(3) —	$s = s_0 + v_0 t + \frac{1}{2}a_0 t^2$
(4) —	$v^2 = v_0^2 + 2a_0(s - s_0)$

Solutions (ii):

new I.C.

$y = h \,, \; V_y = 0$

y-dir^n: (2) $\rightarrow$ $\cancel{V}^{0} = V_0 + a_0 t_2$

$0 = (10 \sin 30^\circ) - 9.81 t_2$

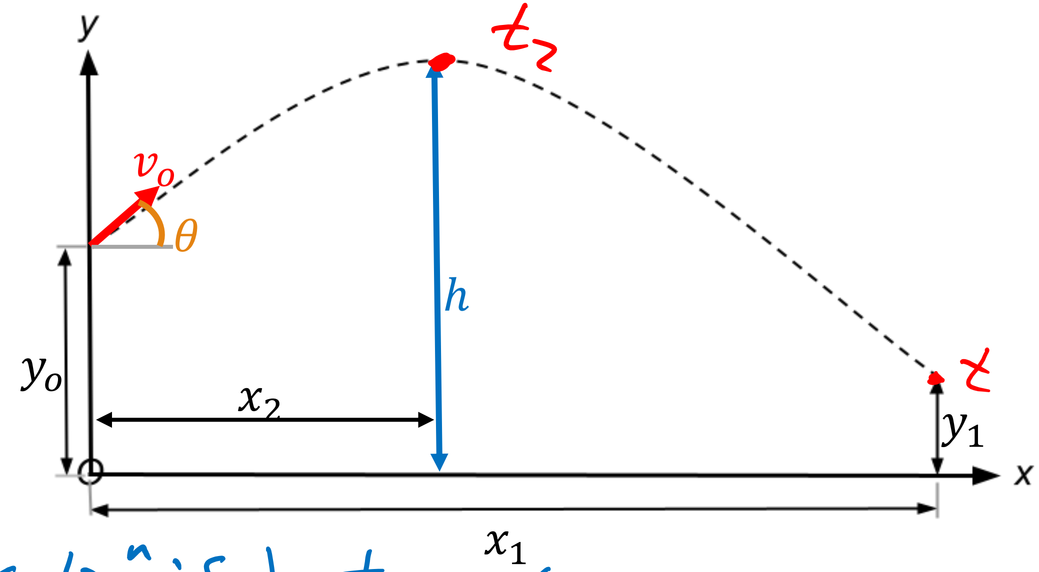
$t_2 = 0.5097 \text{ s}$

Sub t_2 into (3),

$h = y_0 + (V_0 \sin\theta) t_2 - \frac{g}{2} t_2^2$

$h = 10 + (10 \sin 30^\circ)(0.5097) - \frac{9.81}{2}(0.5097)^2$

$\boxed{h = 11.27 m}$ "



x-dir^n: Sub. t_2 — (1)

$x_2 = (V_0 \cos\theta) t_2$

$x_2 = (10 \cos 30^\circ)(0.5097)$

$\boxed{x_2 = 4.41 m}$,,

$y_o = 10m$
$y_1 = 5m$
$v_o = 10 \frac{m}{s}$
$\theta = 30^o$

Special cases	
(1)	(a = 0) $s = s_0 + v_0 t$
(2)	(const a) $v = v_0 + a_0 t$
(3)	$s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$
(4)	$v^2 = v_0^2 + 2a_0(s - s_0)$

Special Cases: Horizontal Acceleration

Special cases	
$(a = 0)$	$s = s_0 + v_0 t$
(const a)	$v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$

Horizontal motion becomes, $\vec{a}_x = -a_x$:

- Velocity

$$v_x = (v_x)_o - a_x t$$

- Position

$$x = x_o + (v_x)_o t - 1/2(a_x t^2)$$

- Velocity-Position

$$v_x^2 = (v_x)_0^2 - 2a_x(x - x_o)$$

Solutions:

old x_1 (w/o a_x) = 14.2 m

$t_1 = 1.64$ s

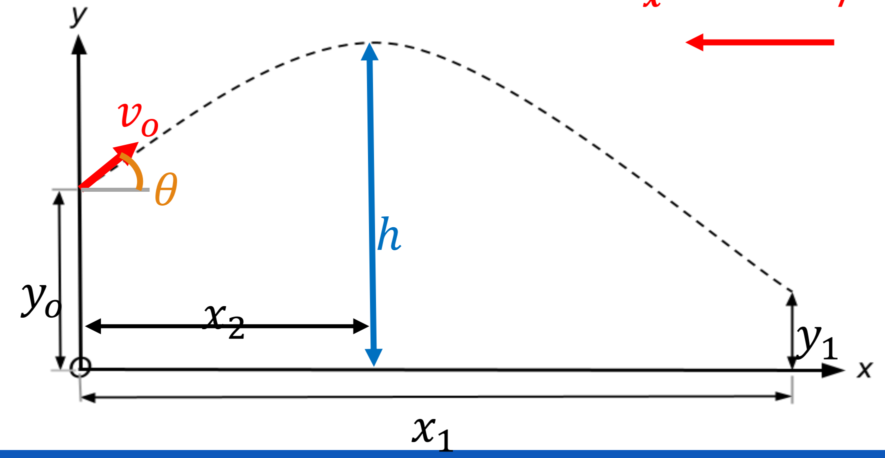
$a_x = -1$

$x = x_0 + v_{x,0} t + \frac{1}{2} a_x t^2$

$x = 0 + (10\cos 35°)(1.64) + (\frac{1}{2})(-1)(1.64^2)$

$x = 12.86$ m < 14.2 m old x_1

$a_x = -1 m/s^2$



In-class Exercise

In-class Exercise [2.4]

Kahoot!

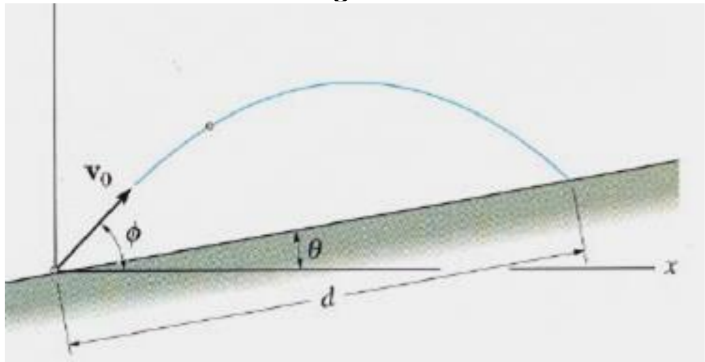
1. A projectile is given an initial velocity v_o at an angle $\emptyset$ above the horizontal. The velocity of the projectile when it hits the slope is _______ the initial velocity v_o.

A) greater than

B) equal to

$\boxed{\text{C) less than}}$

D) None of the above.



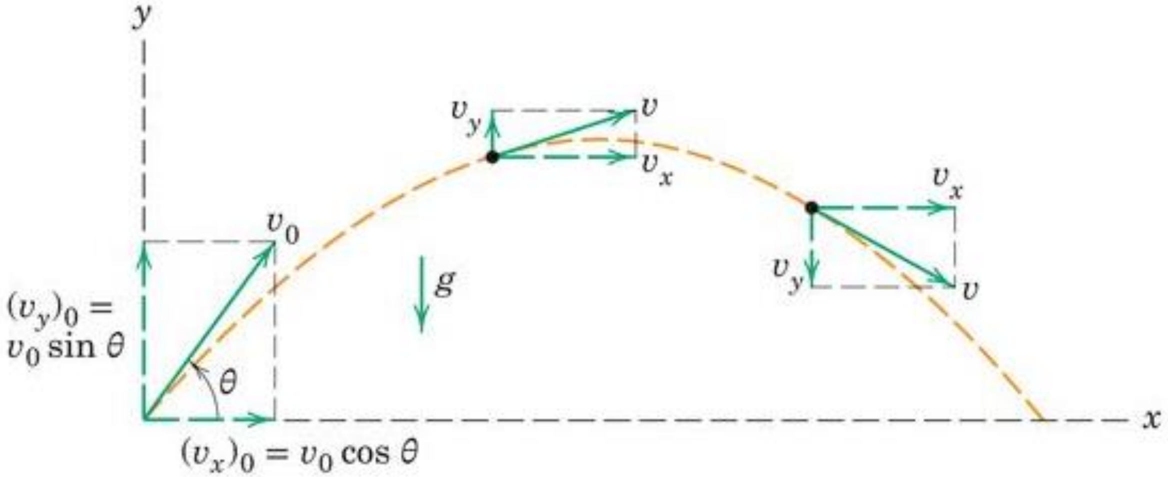
2. A particle has an initial velocity v_o at angle θ with respect to the horizontal. The maximum height it can reach is when

A) $\theta = 30°$

$\boxed{\text{B) } \theta = 90°}$

C) $\theta = 60°$

D) $\theta = 45°$



In-class Exercise [2.4]

Kahoot!

3. The time of flight of a projectile, fired over level ground with initial velocity V_o at angle θ, is equal to

A) $(v_o \sin\theta)/g$

B) $(2v_o \sin\theta)/g$ ✓

C) $(v_o \cos\theta)/g$

D) $(2v_o \cos\theta)/g$

$v_{y,0}$ ✓ v_y ✓

$v_{y,0} = V_0 \sin\theta$

$v_y = -V_0 \sin\theta$

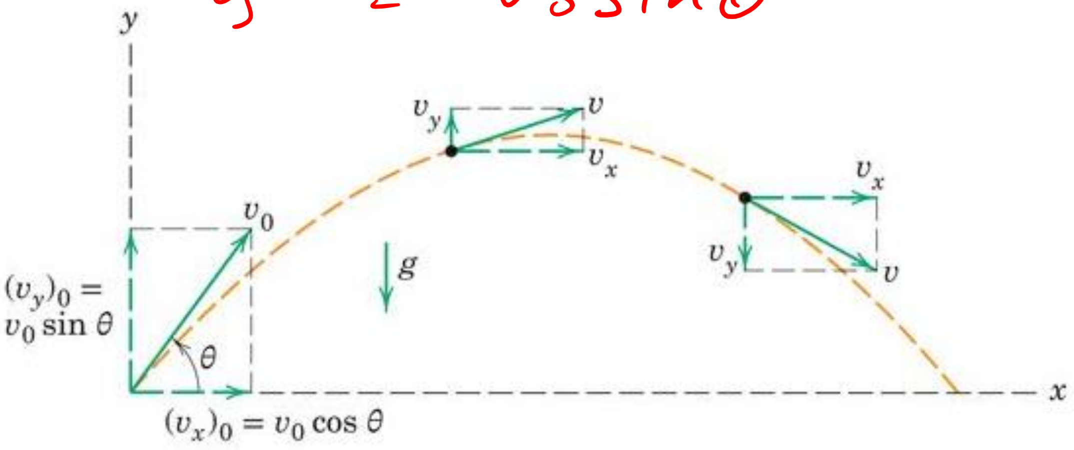
$V = V_0 + a_0 t$

$V_y = V_{y,0} - g t$

$g t = 2 V_{y,0}$

$t = \dfrac{2 V_0 \sin\theta}{g}$

Special cases	
$(a = 0)$	$s = s_0 + v_0 t$
(const a)	$v = v_0 + a_0 t$ $s = s_0 + v_0 t + \frac{1}{2} a_0 t^2$ $v^2 = v_0^2 + 2a_0(s - s_0)$



Practice Questions (from Quercus)

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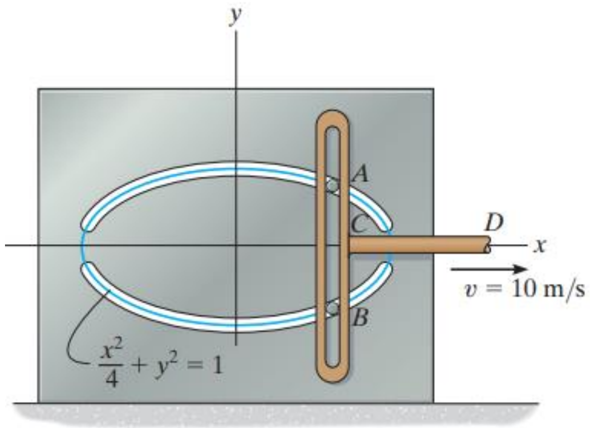
▾ Sample Problems

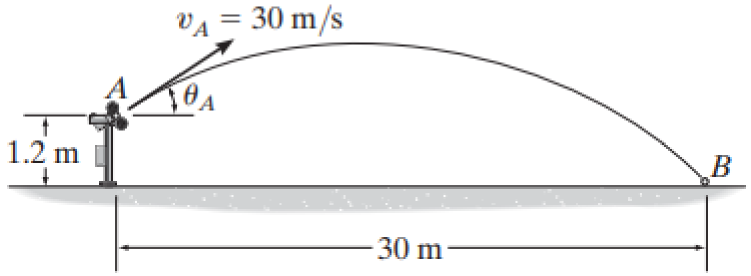
- Chapter 2 - Kinematics of Particles Sample Problems.pdf
- Chapter 3A - Kinetics of Particles_1.pdf
- Chapter 3B - Work Energy Particles.pdf
- Chapter 3C- Kinetics of Particles Momentum.pdf
- Chapter 5- Kinematics of Rigid Bodies Sample Problems.pdf
- Chapter 6A- Kinetics of Rigid Bodies.pdf
- Chapter 6B- Work Energy Rigid Bodies.pdf

12–87.

Pegs A and B are restricted to move in the elliptical slots due to the motion of the slotted link. If the link moves with a constant speed of 10 m/s, determine the magnitude of the velocity and acceleration of peg A when $x = 1$ m.



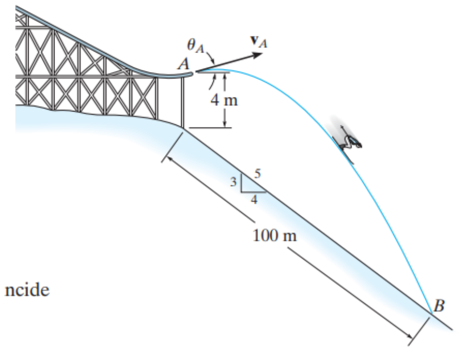
12–91. The pitching machine is adjusted so that the baseball is launched with a speed of $v_A = 30$ m/s. If the ball strikes the ground at B, determine the two possible angles θ_A at which it was launched.



12–101.

It is observed that the skier leaves the ramp A at an angle $\theta_A = 25°$ with the horizontal. If he strikes the ground at B, determine his initial speed v_A and the speed at which he strikes the ground.

ncide



Learning Objective

In this lecture, you will learn to

- Apply the motion equations for analyzing projectile motion
- Analyze the free-flight motion of a projectile particle

Next lecture:
2.5 Normal and Tangential Coordinates (n-t)